

The Stringer bound at $n = 2$: a first finite-sample conservatism theorem, exact counterexamples at low confidence, and a gap in the literature

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Abstract

The Stringer bound (Stringer, 1963) is the standard evaluation of monetary-unit samples in auditing. Its defining property — that its finite-sample coverage is at least the nominal confidence level — has remained unproven for six decades: Bickel (1992) and Pap and van Zuijlen (1996) established asymptotic conservatism for $\alpha \leq 1/2$ and asymptotic anti-conservatism for $\alpha > 1/2$, but no finite-sample proof beyond $n = 1$ exists. We contribute three results. First, we prove the conjecture at $n = 2$: for every taint distribution on $[0, 1]$ and *every* confidence level, the Stringer bound with binomial factors is conservative, and the bound is sharp (the supremum of the failure probability equals α , approached but not attained). To our knowledge this is the first finite-sample proof at any $n \geq 2$ free of hypotheses on the distribution; a second, structurally different proof of the key inequality is sketched. Second, we exhibit the first exact (non-simulation) finite-sample counterexamples at explicit parameters: certified in rational arithmetic, coverage falls below the nominal level at confidence levels 30%–37% for n between 50 and 400, with the shortfall growing in n — while at 50% and above no violation is found, with the searched infimum equal to the nominal level exactly. Third, we refute the only claimed finite-sample certification in the literature (Bimpeh, 2008), which asserted conservatism for all F at $n \leq 11$ for $\alpha = 0.05$ and claimed a proof for $n \leq 2$: its band-containment inequality fails for continuous and atomic distributions alike, and the corrected containment probability can never exceed $1 - \alpha$, so the method could not have certified conservatism at any n — making the present theorem the first correct finite-sample proof at $n = 2$.

1 Introduction

Monetary-unit sampling (MUS) draws n dollar units at random from an audited population; the *taint* of a drawn unit is $T \in [0, 1]$, the overstated fraction of the line item owning that dollar. With taints $t_{(1)} \geq \dots \geq t_{(n)}$ sorted decreasingly, the Stringer bound at nominal confidence $1 - \alpha$ is

$$\text{SB} = p_n(0) + \sum_{j=1}^n (p_n(j) - p_n(j-1)) t_{(j)}, \quad (1)$$

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where $p_n(j)$ is the upper Clopper–Pearson limit for j successes in n trials, i.e. the root of $\sum_{i=0}^j \binom{n}{i} p^i (1-p)^{n-i} = \alpha$, with $p_n(n) = 1$; audit practice frequently substitutes Poisson limits, which dominate the binomial ones. The method’s justification has always been the belief that

$$\mathbb{P}(\text{SB} \geq \mu(F)) \geq 1 - \alpha \quad \text{for all } n \text{ and all } F \text{ on } [0, 1], \quad (2)$$

where $T_1, \dots, T_n$ are i.i.d. F with mean $\mu(F)$ — the standard i.i.d. model in which all published analyses of the bound are set. As the National Research Council’s panel observed, “the formulation of the Stringer bound has never been satisfactorily explained” [9]. What is known: Bickel [3] proved (2) for $n = 1$, gave the lower bound $(1 - \alpha)^{n+1}$ for $n \geq 2$ under conditions on F , and derived the expansion $\text{SB} = \bar{T} + c(F)z_{1-\alpha}n^{-1/2} + o_p(n^{-1/2})$ with $c^2(F) \geq \text{Var}(T)$; Pap and van Zuijlen [11] made the expansion almost sure and concluded that the bound is asymptotically conservative for $\alpha \in (0, \frac{1}{2}]$ and asymptotically *anti*-conservative for $\alpha \in (\frac{1}{2}, 1)$, and in [10] exhibited non-conservatism for uniform-type taints at sub-50% confidence via exact recursions. De Jager, Pap and van Zuijlen [5] proved that on $\{0, 1\}$ -supported taints the bound reduces to Clopper–Pearson (where (2) holds exactly) and that Stringer’s coefficients are minimal with that property. A 2021 survey of confidence bounds for bounded means lists the Stringer bound among “bounds conjectured to have guaranteed coverage,” its coverage for $\alpha < 0.5$ “unknown” [12].

This paper makes three contributions.

(1) *A theorem at $n = 2$* (Section 2): the conjecture (2) holds at $n = 2$ for every F and every $\alpha \in (0, 1)$ — including $\alpha > 1/2$, so the known asymptotic failures are strictly a large- n phenomenon — and it is sharp.

(2) *Exact counterexamples at low confidence* (Section 3): thirty-three certified two-atom counterexamples at confidence levels 30%–37%, $n = 50$ –400, evaluated in exact rational arithmetic with margin certificates; and exhaustive-search evidence that at 90%, 95%, and 50% the infimum over two- and three-atom populations equals the nominal level exactly at every n searched.

(3) *A correction to the record* (Section 4): the only claimed finite-sample certification beyond $n \leq 2$ [4] rests on an index error; we refute its central inequality by hand-checkable counterexamples and prove that its method, corrected, cannot certify conservatism at any n .

2 The theorem at $n = 2$

For $n = 2$ the factors are explicit: $p_2(0) = 1 - \sqrt{\alpha}$, $p_2(1) = \sqrt{1 - \alpha}$, $p_2(2) = 1$. Write

$$\beta = \sqrt{\alpha}, \quad \gamma = \sqrt{1 - \alpha} \quad (\beta^2 + \gamma^2 = 1), \quad A = \beta + \gamma - 1, \quad B = 1 - \gamma,$$

so $A, B > 0$ and $A + B = \beta$.

Theorem 1. *Let $\alpha \in (0, 1)$ and let T_1, T_2 be i.i.d. from any distribution F on $[0, 1]$ with mean μ . Then $\mathbb{P}(\text{SB} \geq \mu) \geq 1 - \alpha$. Moreover $\sup_F \mathbb{P}(\text{SB} < \mu) = \alpha$, and the supremum is not attained.*

2.1 Reduction

Substituting $U_i = 1 - T_i$, $w = 1 - \mu = \mathbb{E}[U]$, $m = \min(U_1, U_2)$, $M = \max(U_1, U_2)$ into (1) and telescoping gives

$$\text{SB} = 1 - (Am + BM), \quad (3)$$

so $\text{SB} \geq \mu \iff Am + BM \leq w$. The theorem is implied by the *wedge inequality*: for U_1, U_2 i.i.d. on $[0, 1]$ with mean $w > 0$,

$$\mathbb{P}(Am + BM < w) \geq \gamma^2. \quad (4)$$

(If $w = 0$ then $U \equiv 0$ a.s. and $\text{SB} = 1 \geq \mu$ surely. The strict form (4) is in fact equivalent to the theorem, by applying the weak form to $(1 - \varepsilon)U$ and letting $\varepsilon \downarrow 0$; only the stated implication is used.)

If $w > \beta$ the event $\{Am + BM \geq w\}$ is empty, since the left side is at most $A + B = \beta$; if $w = \beta$ it is exactly $\{U_1 = U_2 = 1\}$ (as $A, B > 0$), of probability $\mathbb{P}(U = 1)^2 \leq w^2 = \alpha$ by Markov's inequality. **Assume henceforth** $0 < w < \beta$.

2.2 Rectangle bounds

Let $F(x) = \mathbb{P}(U \leq x)$, $G = 1 - F$, and recall $w = \int_0^1 G$. Define

$$u^* = \frac{w}{\beta} \in (0, 1), \quad a_0 = \max\left(0, \frac{w - B}{A}\right), \quad T^\dagger = \min\left(\frac{w}{B}, 1\right),$$

and note $a_0 < u^* < T^\dagger$.

Lemma 2 (Rectangle lemma). *If $0 \leq a \leq b \leq 1$ and $Aa + Bb < w$, then $\mathbb{P}(Am + BM < w) \geq 2F(a)F(b) - F(a)^2$.*

Proof. On $\{U_1 \leq a, U_2 \leq b\} \cup \{U_1 \leq b, U_2 \leq a\}$ we have $m \leq a$, $M \leq b$, hence $Am + BM \leq Aa + Bb < w$ (using $A, B \geq 0$). By inclusion-exclusion (the intersection is $\{U_1 \leq a, U_2 \leq a\}$ since $a \leq b$) the union has probability $2F(a)F(b) - F(a)^2$. $\square$

Assume for contradiction that (4) fails. Then

$$2F(a)F(b) - F(a)^2 < \gamma^2 \quad \text{whenever } 0 \leq a \leq b \leq 1, Aa + Bb < w. \quad (\text{H})$$

Three families of admissible rectangles give pointwise bounds on G :

(i) *Squares.* For $t < u^*$, $(a, b) = (t, t)$ gives $F(t)^2 < \gamma^2$, i.e. $G(t) > 1 - \gamma = B$ for all $t \in [0, u^*)$.

(ii) *Full-height rectangles* (nonvacuous iff $w > B$). For $a < a_0$, $(a, 1)$ gives $1 - G(a)^2 < \gamma^2$, i.e. $G(a) > \beta$ on $[0, a_0)$.

(iii) *Boundary pairs.* Let $h(g) = \frac{\beta^2 - g^2}{2(1 - g)}$ on $[0, 1]$. For $b \in (u^*, T^\dagger)$ set $\bar{a}(b) = (w - Bb)/A \in (a_0, u^*)$. Every $a' < \bar{a}(b)$ is admissible with b ; letting $a' \uparrow \bar{a}(b)$ and writing $g = G(\bar{a}(b)^-) < 1$, inequality (H) rearranges to

$$G(b) \geq h(G(\bar{a}(b)^-)) \quad \text{for all } b \in (u^*, T^\dagger)$$

(the numerator algebra: $2(1 - g) - \gamma^2 - (1 - g)^2 = \beta^2 - g^2$). If $g = 1$ the bound $G(b) \geq 0$ is used instead, consistent with the convention $h_+(1) = 0$ below.

2.3 Potential function and budget

Let $h_+ = \max(h, 0)$, extended continuously by $h_+(1) = 0$, and define

$$\tilde{k}(g) = g + \frac{A}{B} h_+(g), \quad g \in (B, 1].$$

Lemma 3. $\tilde{k}(g) \geq \beta$ on $(B, 1]$, with equality only at $g = \beta$.

Proof. Since $\beta^2 - B^2 = \alpha - (1 - \gamma)^2 = 2\gamma(1 - \gamma)$, we get $h(B) = B$, hence $\tilde{k}(B^+) \rightarrow B + \frac{A}{B}B = A + B = \beta$; also $h(\beta) = 0$ so $\tilde{k}(\beta) = \beta$. Differentiating, $h'(g) = \frac{(1-g)^2 - \gamma^2}{2(1-g)^2}$, so h has its unique critical point at $g = B$ (where $(1 - g)^2 = \gamma^2$) and decreases on $(B, 1)$; hence $h \geq 0$ on $[B, \beta]$ (it decreases from $B > 0$ to 0) and $h < 0$ on $(\beta, 1)$. Further $h''(g) = -\gamma^2/(1 - g)^3 < 0$; h , and therefore $\tilde{k} = g + \frac{A}{B}h$, is strictly concave on $[B, \beta]$. A strictly concave function with equal endpoint values β satisfies $\tilde{k} \geq \beta$ on $[B, \beta]$, strictly on the interior. On $(\beta, 1]$, $h_+ = 0$ and $\tilde{k}(g) = g > \beta$. $\square$

Proof of Theorem 1. Assume (H). The substitution $t = b(a) = (w - Aa)/B$ maps $[a_0, u^*]$ decreasingly onto $[u^*, T^\dagger]$ with $|dt| = (A/B) da$ (endpoints: $b(u^*) = u^*$; $b(a_0) = 1 = T^\dagger$ if $w \geq B$, $b(0) = w/B = T^\dagger$ if $w \leq B$). Using $w = \int_0^1 G \geq \int_0^{a_0} G + \int_{a_0}^{u^*} G + \int_{u^*}^{T^\dagger} G$, bound the tail by (iii) (the left-limit $G(a^-)$ agrees with $G(a)$ for all but countably many a , so the integrals coincide):

$$\int_{u^*}^{T^\dagger} G = \frac{A}{B} \int_{a_0}^{u^*} G(b(a)) da \geq \frac{A}{B} \int_{a_0}^{u^*} h_+(G(a)) da.$$

By (i), $G > B$ pointwise on $[a_0, u^*)$, so Lemma 3 applies under the integral:

$$w \geq \int_0^{a_0} G + \int_{a_0}^{u^*} \tilde{k}(G(a)) da \geq \beta a_0 + \beta(u^* - a_0) = \beta u^* = w,$$

where $\int_0^{a_0} G \geq \beta a_0$ by (ii). Equality is forced throughout; we exhibit a strict inequality in every case:

- If $a_0 > 0$: (ii) is pointwise strict on $[0, a_0)$, so $\int_0^{a_0} G > \beta a_0$.
- If $a_0 = 0$ and $G \neq \beta$ on a positive-measure subset of $(0, u^*)$: there $\tilde{k}(G) > \beta$ strictly (Lemma 3), so $\int_0^{u^*} \tilde{k}(G) > \beta u^*$.
- If $a_0 = 0$ and $G = \beta$ a.e. on $(0, u^*)$: for each $b \in (u^*, T^\dagger)$ choose $a' < \bar{a}(b)$ with $G(a') = \beta$ (possible, as such a' have full measure in a nonempty interval); the *strict* inequality (H) at the admissible corner (a', b) reads $(1 - \beta)(1 + \beta - 2G(b)) < \gamma^2 = (1 - \beta)(1 + \beta)$, forcing $G(b) > 0$ for every $b \in (u^*, T^\dagger)$; hence $\int_{u^*}^{T^\dagger} G > 0$ and $w \geq \int_0^{u^*} G + \int_{u^*}^{T^\dagger} G > \beta u^* = w$.

Each case contradicts $w = w$; so (H) is impossible, proving (4) and the coverage statement.

Sharpness. Two-point distributions $F = (1 - q)\delta_0 + q\delta_u$ give failure probability q^2 when $q < \beta$ (the pair (u, u) , since $\beta u > qu = w$) and additionally the mixed pairs when $q < B$ (since $Bu > qu$), totalling $1 - (1 - q)^2$; letting $q \uparrow \beta$ and $q \uparrow B$ respectively yields failure probabilities approaching $\beta^2 = \alpha$ and $1 - \gamma^2 = \alpha$. Hence $\sup_F \mathbb{P}(\text{SB} < \mu) = \alpha$.

Non-attainment. If some F attained it, the weak form of (H) would hold, and rerunning the budget with weak inequalities forces $G = \beta$ a.e. on $(0, u^*)$ (by strictness of $\tilde{k}$ off β) and $G = 0$ a.e. beyond u^* ; right-continuous monotonicity then pins $F = (1 - \beta)\delta_0 + \beta\delta_{u^*}$, whose largest value of $Am + BM$ is $\beta u^* = w$ exactly, so $\mathbb{P}(Am + BM > w) = 0 \neq \alpha$ — a contradiction. $\square$

Remark 4 (Relation to known inequalities). The equal-weights case of (4) — bounding $\mathbb{P}(X_1 + X_2 \geq t)$ for nonnegative i.i.d. variables with given mean — is the classical sharp inequality of Hoeffding and Shrikhande [7] (extended to general k in [8]); the wedge inequality is its order-statistic-weighted analogue, with genuinely unequal weights $A \neq B$ for all $\alpha \in (0, 1)$. The sharp-bound literature for order statistics and L-statistics under moment conditions (see [2]) concerns expectations, or distribution functions of single order statistics, not mean-constrained tail probabilities of weighted combinations; we are not aware of a prior result implying (4). Separately, the valid-mean-bound literature offers an instructive comparison: Gaffke’s bound [6], whose finite-sample validity for independent variables was proved only recently [14], is pointwise dominated by the $n = 2$ Stringer bound at $\alpha = 0.05$ (numerically: minimum gap 0, attained exactly on the two extremal families — equal taints, and one taint equal to 1), so at standard confidence levels conservatism at $n = 2$ would alternatively follow from [14] *given* a proof of that domination; but the domination fails at $\alpha > 1/2$ (e.g. by -0.032 at $\alpha = 0.7$), where Theorem 1 still holds, so the present proof is strictly stronger and self-contained. Anderson’s bound [1] does not dominate or get dominated similarly.

Remark 5 (Second proof). An independent verification produced a structurally different proof of (4): decompose the failure probability as $g^2 + 2 \int_{[0, u^*)} G(h(u)) dF(u)$ with $g = G(u^*)$, bound the tail survival by mass counting, majorize linearly in the high-part mean, and check two endpoints. Its closing inequality, $J(g) = A(\beta + g) - 2B(1 - g) \geq 0$ on $[B, \beta]$ with $J(B) = \beta^2 + \gamma^2 - 1 = 0$, is algebraically equivalent to Lemma 3. Two different arguments hinging on the same tight inequality, with the same equality points $g \in \{B, \beta\}$ matching the two extremal families, corroborate the sharpness structure.

Corollary 6. *For $n = 2$ the Stringer bound with Poisson (AICPA) factors is conservative for every F and every α for which the Poisson factors dominate the binomial ones; the domination is verified with rigorous enclosures for standard levels in the repository.*

Proposition 7 (Single nonzero taint value). *For every n , α , and F supported on $\{0, v\}$ with $v \in (0, 1]$: $\mathbb{P}(\text{SB} \geq \mu) \geq 1 - \alpha$.*

Proof. With K the number of nonzero taints, $K \sim \text{Bin}(n, q)$, $\text{SB} = (1 - v)p_n(0) + v p_n(K) \geq v p_n(K)$ and $\mu = qv$; on the Clopper–Pearson coverage event $\{p_n(K) \geq q\}$, of probability at least $1 - \alpha$, we get $\text{SB} \geq vq = \mu$. $\square$

3 Exact counterexamples at low confidence

Theorem 1 covers every confidence level, so the asymptotic anti-conservatism at $\alpha > 1/2$ [11] must manifest only at larger n . We locate it precisely. All claims below are certified: the taint distribution’s parameters are exact dyadic rationals, coverage is computed as an exact rational by full multinomial enumeration, and a margin certificate verifies that no comparison

$SB \geq \mu$ was decided within the numerical error of the confidence factors (which are enclosed in bisection brackets of width 10^{-40}).

confidence	n	worst certified coverage	certified examples
30%	50	0.29888	3
32%	100	0.30956	9
35%	200	0.33668	1
35%	400	0.32630	1
37%	200	0.36079	6
37%	400	0.35508	13

Table 1: Certified finite-sample violations (binomial factors, two-atom taint distributions). The shortfall grows with n at fixed confidence, and the onset n grows as confidence approaches the asymptotic threshold 50%.

A representative certificate: at nominal confidence 30% ($\alpha = 0.7$), $n = 50$, taints $v_1 \approx 0.99$, $v_2 \approx 0.4655$ with probabilities $q_1 \approx 0.246$, $q_2 \approx 0.592$ (exact dyadic values in the repository), the exact coverage is $0.29888\dots < 0.30$, with minimum margin $\min |SB - \mu| \approx 2 \times 10^{-17}$ against a factor-error bound of 5×10^{-39} . These are, to our knowledge, the first exact counterexamples at explicit parameters in the literature; [10] established existence for uniform-type taints via recursions but the present examples are two-atom, elementary to re-verify, and map the failure boundary.

At and above 50% confidence the picture reverses. Grid-plus-optimizer searches over two-atom populations ($n \leq 100$ at 95% and 90%; $n \leq 300$ at 50%) and three-atom populations ($n \leq 25$ at 95%) find *no* violation; at every n the screened infimum equals the nominal level to 10^{-10} or better, approached as the top taint $v_1 \rightarrow 1$ — consistent with Proposition 7 and with the $\{0, 1\}$ -support minimality of [5], and evidence that the conjecture, if true, is exactly tight. For the Poisson factors used in practice the worst-case coverage stays strictly above nominal, e.g. at 95%: 0.9716 ($n = 10$), 0.9573 ($n = 50$), 0.9549 ($n = 100$) — a finite-sample cushion that erodes as n grows.

4 The gap in the claimed certification

Bimpeh [4] writes the Stringer bound as $SB = \int_0^1 [1 - \hat{F}_{n,L}(t)] dt$, where $\hat{F}_{n,L}$ is the stepwise band of lower Clopper–Pearson limits $q_n(i-1)$ on $[t_{i-1:n}, t_{i:n})$, and derives

$$CP \geq \bar{P}_n = \mathbb{P}(U_{i:n} \leq p_n(i), 1 \leq i \leq n), \quad (5)$$

computing $\bar{P}_n$ by Bolshev’s recursion and concluding that (2) holds for all F when $\bar{P}_n \geq 1 - \alpha$ — e.g. $n \leq 11$ at $\alpha = 0.05$ — and, since $\bar{P}_2 = 2\sqrt{1 - \alpha} - (1 - \alpha) \geq 1 - \alpha$ for all α , claiming in particular a proof of (2) for $n \leq 2$. This is the only finite-sample certification claimed anywhere in the literature; to our knowledge it has never been cited critically, so the error identified below has gone unnoticed since 2008.

Proposition 8. *Inequality (5) is false, for continuous and for atomic F .*

Proof. Let $\alpha = 0.05$ and let F put mass 0.99 uniformly on $(0.99, 1)$ and 0.01 uniformly on $(0, 0.01)$, so $\mu = 0.9851$. For $n = 1$: $\text{SB} = 0.95 + 0.05 T_1$ and $\bar{P}_1 = \mathbb{P}(U \leq p_1(1)) = 1$; but $\text{SB} < \mu \iff T_1 < 0.702$, an event of probability exactly 0.01, so $\text{CP} = 0.99 < 1 = \bar{P}_1$. For $n = 2$: $\bar{P}_2 = 2\sqrt{0.95} - 0.95 = 0.99936$, while the pair (one high taint, one low) fails, giving $\text{CP} = 1 - 2(0.99)(0.01) - 0.01^2 = 0.9801 < \bar{P}_2$. An atomic example at $n = 5$: F with $\mathbb{P}(T = 1) = \frac{1}{2}$, $\mathbb{P}(T = \frac{1}{10}) = \mathbb{P}(T = 0) = \frac{1}{4}$ has exact coverage $\frac{31}{32} = 0.96875 < \bar{P}_5 = 0.98746$ (the failure event is precisely the 2^{-5} -probability event of drawing no taint-1 unit). $\square$

The error is an index shift: pointwise domination $F \geq \hat{F}_{n,L}$ on $[0, 1]$ is equivalent to $F(t_{i:n}) \geq q_n(i)$ for all i — the band’s value *at* the jump points — whereas [4] substitutes the left limits $q_n(i - 1)$, dropping in particular the top constraint $F(t_{n:n}) \geq q_n(n) = \alpha^{1/n}$. Moreover the method cannot be repaired:

Proposition 9. *For every continuous F and every n , the corrected containment probability satisfies $\mathbb{P}(F(t_{i:n}) \geq q_n(i) \ \forall i) = \mathbb{P}(U_{i:n} \geq q_n(i) \ \forall i) \leq \mathbb{P}(U_{n:n} \geq \alpha^{1/n}) = 1 - \alpha$.*

Thus the pointwise-band route yields at most $1 - \alpha$ and can certify conservatism at no n : if (2) is true, it is true because the integral inequality $\int \hat{F}_{n,L} \leq \int F$ holds on a strictly larger event than pointwise domination. (The values of $\bar{P}_n$ themselves, and Bimpeh’s Table 5.1, reproduce exactly; they are simply not coverage bounds.) In particular the $n = 1, 2$ counterexamples of Proposition 8 refute (5) at precisely the sample sizes for which [4] claimed an analytic proof, so that claim is invalid and Theorem 1 is, to our knowledge, the first correct finite-sample proof at $n = 2$. The proven record therefore reduces to $n = 1$ [3], $\{0, 1\}$ supports [5], single-value supports (Proposition 7), and now $n = 2$ (Theorem 1).

5 Discussion

Three observations orient future work. First, the reformulation behind (3) generalizes: with $u_{(1)} \leq \dots \leq u_{(n)}$ the increasing order statistics of $U_i = 1 - T_i$,

$$\text{SB} = 1 - \sum_{j=1}^n e_j u_{(j)}, \quad e_j = p_n(j) - p_n(j - 1),$$

so (2) is a family of weighted order-statistic exceedance inequalities, with all audit content absorbed into the decreasing weights e_j . Second, the $n = 2$ proof’s architecture — rectangle lower bounds on the covering event, a survival-integral budget, and a concave potential with equal endpoint values — does not obviously depend on $n = 2$ except through the explicit factor algebra; the $n \geq 3$ factors lack closed forms, so the endpoint identities (Lemma 3) would need structural rather than algebraic proofs. Third, the empirical sharpness at every n and the certified failures below 50% confidence suggest the true statement is exactly (2) for $\alpha \leq 1/2$, with the infimum equal to $1 - \alpha$ at every n — a conjecture now precise enough to attack.

For practice: the binomial Stringer bound has no worst-case safety margin at all (95% means exactly 95% against adversarial populations), the Poisson tables retain a quantifiable but eroding cushion, and evaluations below 50% confidence are provably anti-conservative at audit-relevant sample sizes.

Reproducibility

Every numerical claim is backed by machine-verifiable artifacts in the accompanying repository: exact rational coverage computations with margin certificates, rigorous factor enclosures, the search logs, a four-layer verification of Theorem 1 (reduction identity, symbolic identities, potential bound, and a $\sim 200,000$ -distribution stress test), and records of two independent adversarial reviews of the proof, one containing the alternative argument of Remark 5.

AI disclosure

Anthropic Claude was used for mathematical exploration, generation and adversarial criticism of candidate arguments, code, literature search, and manuscript drafting, directed and reviewed by the author, who accepts responsibility for the contents. AI review is not a substitute for independent human verification; the proofs are written to be checked by hand.

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